

KUNAL MATHUR

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GitHub: github.com/kunal202426 | Portfolio: portfolio-kunal26.vercel.app

SUMMARY

Full Stack SDE intern at YES Bank shipping production features on a high-throughput trading platform processing 20,000+ daily trades; national competition winner at IIT Madras Global Hyperloop competition and Flipkart GRiD 7.0. Specialises in scalable React and Node.js systems, backend microservices, and PostgreSQL at production scale.

EDUCATION

Vellore Institute of Technology
B.Tech, Computer Science & Engineering

2022 – 2026
Vellore, TN

EXPERIENCE

YES Securities (Yes Bank)

Jan 2026 – Jun 2026

Full Stack Developer Intern

Bengaluru, India

- Shipped 5+ production features for the OMNI trading application (React, Node.js, RESTful APIs) in Agile sprints, serving 20,000+ daily trades; refactored backend microservices cutting average API latency by ~25%.
- Resolved N+1 query bottlenecks and redesigned PostgreSQL schemas reducing critical query execution time by ~35%; streamlined CI/CD pipeline across 3 backend services and delivered an end-to-end ML forecasting pipeline in production.

PROJECTS

Distributed Systems Design Simulator

Oct 2025 – Nov 2025

- Built a platform to design, visualise, and stress-test distributed architectures in real time, modeled failure modes via M/M/1 queueing theory, chaos engineering, and probabilistic fault injection.

CityFlow: Real-Time 3D City Simulation & Analytics

Feb 2026 – Apr 2026

- Urban logistics simulation modeling 5 demand variables (fleet size, surge pricing, weather, demand zones, time-of-day) with 4 live KPIs tracked on a real-time interactive 3D dashboard.

Hybrid Stock Price Prediction System

Jan 2026 – April 2026

- End-to-end ML forecasting pipeline (LSTM + LightGBM + FinBERT ensemble) reducing MAE by ~18% over single-model baselines; deployed as a production internship deliverable at YES Bank.

SKILLS

Languages: Python, JavaScript (ES6+), Java, SQL

Frontend: React.js, Three.js, ReactFlow, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Microservices

Databases: PostgreSQL, MongoDB, Firebase

Tools and Cloud: Git, GitHub, AWS, Vercel, Figma, CI/CD

ML and AI: TensorFlow, Scikit-learn, LSTM, LightGBM, FinBERT, ETL Pipelines

ACHIEVEMENTS

Winner – Global Hyperloop Competition (Asia's First), IIT Madras

2025

6 awards across engineering, design, and technical presentation out of 30+ competing teams nationally, representing VIT Vellore.

National Semi-Finalist in Flipkart GRiD 7.0

2025

Advanced to national semi-finals among thousands of engineering participants in a real-world software engineering challenge.

CERTIFICATIONS

Data Analytics — *GeeksforGeeks, 2026* · 98% score (874/895) – Python, Pandas, NumPy, SQL, Power BI, Tableau

Blockchain Developer — *IBM Career Education, 2025* · Smart contracts, Hyperledger Fabric, DApp architecture

LEADERSHIP AND EXTRACURRICULARS

Team Levitate Hyperloop

2024 – 2025

AI and Software Lead

Vellore, India

- Led full-stack development of team website with real-time 3D Pod simulations (Three.js, React); built live hardware telemetry pipelines achieving sub-50ms data refresh, contributing to 6 awards at IIT Madras in Global Hyperloop competition.